

Name:

BUID:

TF Name:

Section:

MOLECULAR SPECTROSCOPY WORKSHOP LAB WORKSHEET

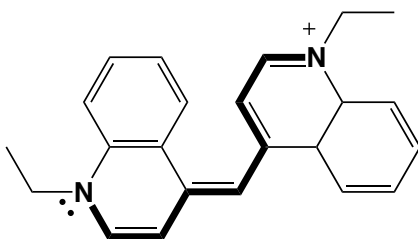
Complete this through at least Part 3 of this worksheet during the lab. Include all your work and answers in the provided space for each question. Finish Part 4, then scan and upload it to Gradescope before the due date posted there.

During the workshop, make sure you:

- ☐ Get feedback from a lab instructor on the techniques for using a spectrophotometer
- ☐ Practice using the Cary-60 UV-Vis for one of the samples from the workshop

PART 1 – STUDYING THE CONCENTRATION OF A DYE USING BEER'S LAW

Large molecules produce brightly colored solutions because of the conjugated molecular π -systems that absorb light in the visible and ultraviolet regions of the electromagnetic spectrum. For example, 1,1'-diethyl-4,4'-cyanine iodide (shown below) is a conjugated molecule that has a strong absorption in the visible region.



As derived in the lab background, the absorbance of a solution of light-absorbing species is directly related to the concentration of the solution, the distance the light travels through the sample (i.e., the pathlength), and the molar absorptivity of the species. We will now apply Beer's Law ($A = \epsilon bc$) to determine the concentration of a dye solution with unknown concentration.

- (a) You will be provided with a solution of 1,1'-diethyl-4,4'-cyanine iodide ($M = 454.35 \frac{\text{g}}{\text{mol}}$) with a concentration of approximately 2.5 mg of dye in 500. mL of solution. Calculate the molar concentration (in moles/L) of the dye in solution, based on the actual concentration (in mg/mL) reported for the solution.

Name:

BUID:

- (b) Record the spectrum of 1,1'-diethyl-4,4'-cyanine iodide. Sketch the visible region of the spectrum (380-900 nm) and note the absorbance of the solution at the maximum absorbance wavelength.
- (c) Use the absorbance at λ_{max} and the concentration from (a) to compute the molar absorptivity of 1,1'-diethyl-4,4'-cyanine iodide at that wavelength. Make sure to include the appropriate units. *Make sure to use the correct pathlength.*
- (d) Another solution of 1,1'-diethyl-4,4'-cyanine iodide, with unknown concentration, has also been provided to you by your instructor. Record the spectrum. Use Beer's Law to determine the concentration (in moles/L) of the unknown solution.
- (e) Using what you know about Beer's Law, why did you make all your measurements at a single wavelength? Explain in 1 sentence.
- (f) What might have been the effect on your UV-Vis absorption spectrum if you did not measure an appropriate blank? Explain in 1 sentence.

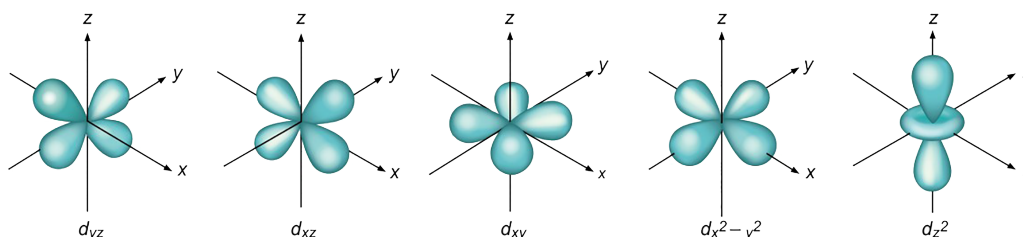
Name:

BUID:

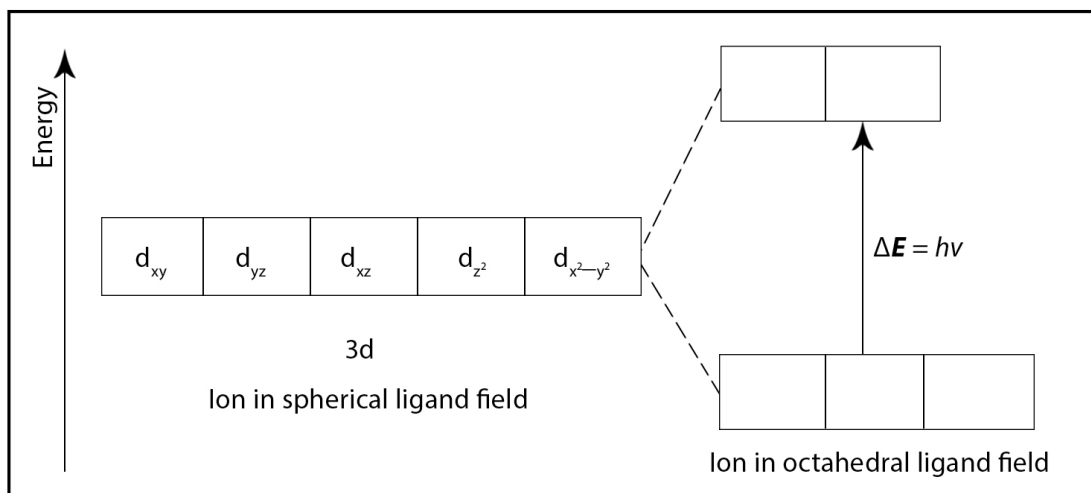
PART 2 – SPECTRA OF COPPER(II) ION COMPLEXES

In this part of the workshop we will explore crystal field theory, and examine the phenomenon of light absorption by metal complex ions. Additionally, we will explore how the nature of the ligand and metal affects the complexes. For the hydrated copper(II) ion, the absorption occurs at the red end of the spectrum, which leads to a complex that appears blue in solution.

- (a) Before the ligands approach the metal ion, the five 3d orbitals are degenerate (same energy). As the electrons of the ligands approach the metal ion, repulsion between ligands and 3d orbitals will cause the energies of the orbitals to split. Circle the two 3d orbitals that will have the greatest overlap with ligands in an octahedral geometry (approaching the ion along the axes' directions).



- (b) The orbitals with the greatest repulsion will increase in energy while the remaining orbitals will be stabilized (go to lower energy). Complete the diagram below by adding the correct orbital designations ($3d_{xy}$, $3d_{xz}$, etc.) to the boxes in the right-hand side of the diagram based on your answer to the previous question.



Splitting of the five degenerate metal ion 3d orbitals in an octahedral ligand field.

Name:

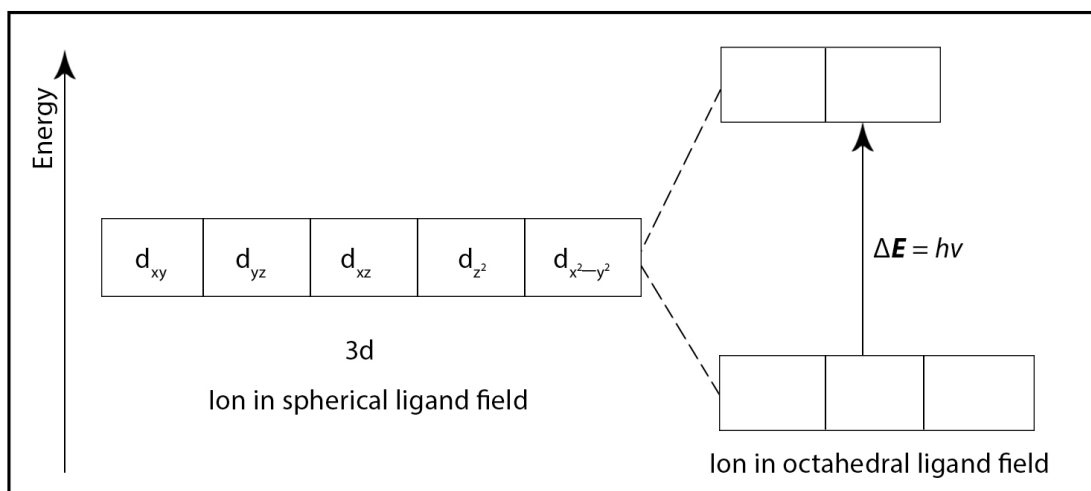
BUID:

- (c) The energy gap between the two levels of d orbitals is the crystal field stabilization energy, ΔE . What is the size of the energy gap (in kJ/mol) that leads to the absorption of a light with $\lambda = 500.$ nm?

- (d) What is the ground-state electronic configuration of the Cu^{2+} ion (without any ligands)?

- (e) Fill-in the orbital diagram below with electrons (use up and down arrows to symbolize electrons) for Cu^{2+} octahedral ions in the ground state.

What transition occurs when light is absorbed by Cu^{2+} in an octahedral ligand field? (i.e. which orbitals are involved? e.g. the first line in the hydrogen atom absorption spectrum might be a “ $2s \rightarrow 3p$ ” transition)



Splitting of the five degenerate metal ion 3d orbitals in an octahedral ligand field. Fill in the electrons for the ground state electronic configuration of copper(II).

Name:

BUID:

- (f) Collect a 5 mL sample of 0.02 M solution of $\text{Cu}(\text{NO}_3)_2$. What is the color of your solution? When $\text{Cu}(\text{NO}_3)_2$ dissolves in water, a complex ion forms. What complex ion is responsible for this color? In what color region of the spectrum do you expect this complex to absorb light?

- (g) Record the UV/Vis spectrum of the $\text{Cu}(\text{NO}_3)_2$ solution, from 400 - 900 nm, using a benchtop spectrophotometer and sketch the spectrum here.

Given that the solution is colored, we expect an absorption in the visible region of the spectrum. Locate the absorption in the **visible region** of the spectrum: what is the wavelength and absorbance at which the most light is absorbed?

- (h) When ammonia is added to the solution, the $[\text{Cu}(\text{NH}_3)_4(\text{H}_2\text{O})_2]^{2+}$ complex ion will form. Add 3-5 drops of concentrated NH_3 to your cuvette with the $\text{Cu}(\text{NO}_3)_2$ solution and mix it well. How does this solution differ in appearance from the stock $\text{Cu}(\text{NO}_3)_2$ solution? (If a cloudy solid forms, consult with your lab instructor before proceeding.)

Name:

BUID:

- (i) Record the UV/Vis spectrum of the $[\text{Cu}(\text{NH}_3)_4(\text{H}_2\text{O})_2]^{2+}$ solution from 400-900 nm using the benchtop spectrophotometer. Sketch the spectrum. Locate the absorption in the **visible region** of the spectrum: what is the wavelength at which the most light is absorbed? Record the absorbance.
- (j) Compare the crystal field splitting energies of the two copper complexes. Which is the weaker field ligand, NH_3 or H_2O (i.e., which ligand causes the smallest splitting between the d orbitals)? Explain in 1 - 2 sentences.
- (k) What is the electronic configuration of the Zn^{2+} ion?
- (l) What d-d electronic transition(s), if any, will happen in octahedral $\text{Zn}^{2+}(\text{aq})$ ions?
- (m) Based on your answer to the previous question, what color do you expect to see for a solution of $\text{Zn}^{2+}(\text{aq})$ ions? Explain in 1 sentence.

Name: _____

BUID:

PART 3 – ABSORPTION, EMISSION, AND FLUORESCENCE OF FLUORESCEIN SOLUTIONS

In normal absorption spectroscopy, emission occurs when the light that is absorbed by a species (atom or molecule) is subsequently released as light. As we've seen, a substance gains energy when it absorbs light. If no other processes take place, the energy given off during the emission of light from that substance will be the same as the energy of the light absorbed. As a result, the emitted light will be the same wavelength as the absorbed light.

Recall, however, that none of the molecules we've studied up until now in this workshop have emitted any light (we've only seen transmitted light from the molecules, nothing emitted). The reason for this is that, in general, the process of atoms and molecules emitting light is slow. As a result, molecules – which can release energy in other ways – may not actually undergo emission. In many cases, light that is absorbed by molecules is released as heat through a process called *relaxation* (i.e., the transfer of energy as vibrations to other molecules). Since atoms, and atomic ions, don't have vibrational or rotational modes, they can only release the energy through emission.

When a molecule *fluoresces*, or phosphoresces, the light emitted by the molecule is lower in energy than the light that the molecule originally absorbed. This process involves a small amount of energy being released as heat (through relaxation) before the emission occurs.

- [illegible]

Name:

BUID:

- (c) Prepare your 10 μM fluorescein solution. Record the absorption spectrum of fluorescein and sketch the spectrum. What is the wavelength (in nm) and energy (in kJ/mol) of the visible light being absorbed by fluorescein? What color visible light is absorbed?
- (d) Based on the color of the light that is absorbed, what color should a fluorescein solution appear? Explain in 1 sentence.
- (e) Remove the cuvette from the spectrometer and note the appearance of the solution. Also use a handheld UV light (see your instructor) to excite the sample. What do you observe? Is the light being emitted higher or lower energy than the light being absorbed?
- (f) What is the energy (in kJ/mol) of the photon being emitted in the fluorescence transition (510 nm)?
- (g) What is the difference in energy between the photon absorbed and emitted by fluorescein? Where did that energy go?

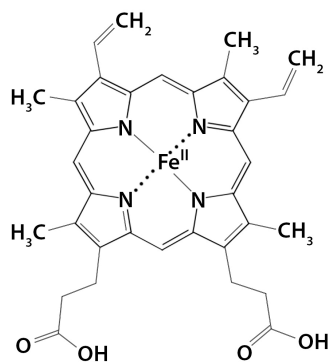
Name:

BUID:

- (h) Draw an energy change diagram that shows the absorption and emission that is happening in fluorescein.

PART 4 – FOLLOW-UP: LIGHT ABSORPTION BY METAL-CONTAINING BIOLOGICAL MOLECULES

Hemoglobin is able to bind molecular oxygen because of a heme group (see below), a cyclic porphyrin molecule that complexes an iron ion. In addition to the four nitrogen ligands from the porphyrin, the protein donates a fifth ligand to the iron ion. The final ligand is either a water molecule or molecular oxygen. When an oxygen molecule coordinates to the iron in hemoglobin, the iron(II) ion is oxidized to iron(III).



Heme molecule: iron ion with the conjugated π system as a tetradentate (4 ligand sites) ligand.

Based on the material that we've learned there are two possible explanations for the bright colors (and color changes) that hemoglobin exhibits: either the bright colors are due to absorption by the iron complex ion at the center of the molecule, or they are due to the conjugated π -system of the porphyrin.

Name:

BUID:

- (a) Beer's law relates the concentration (c) of a species in solution to the absorbance of light (A) due to that species. In a solution of hemoglobin, the concentration of the metal ion and the porphyrin are the same (*why?*). Which variable in the equation for Beer's Law is responsible for the vibrant color of the solution? Explain in 1 sentence.
- (b) Calculate the molar absorptivity (at λ_{max}) for the $[\text{Cu}(\text{NH}_3)_4(\text{H}_2\text{O})_2]^{2+}$ solution that you prepared in Part 3 of the workshop.
- (c) Assume that the molar absorptivities that you computed for your copper complex (Part 2) and the cyanine dye (Part 1) are typical for those classes of molecules – i.e., that the molar absorptivity you computed for $[\text{Cu}(\text{NH}_3)_4(\text{H}_2\text{O})_2]^{2+}$ is typical of metal-ion complexes and the molar absorptivity you computed for 1,1'-diethyl-4,4'-cyanine is typical of conjugated systems. Based on this assumption, what is responsible for the bright colors associated with hemoglobin? Explain your reasoning.